

Hybrid Three-Dimensional LTD/CFF Representations

Implementation and theory note

April 2026

Abstract

This note describes the derivative-free three-dimensional representation implemented in `hybrid3d.py`. The construction combines loop-tree duality (LTD) with the causal/CFF representation in order to handle repeated propagators while keeping the numerator as a sampled black box. It also describes the bounded-energy-degree extension used when edge-momentum representation (EMR) numerators contain powers above the affine CFF class.

The document is organized as follows. Chapter 1 gives the mathematical construction, with all dependencies made explicit. Chapter 2 explains how the Python implementation and the oriented JSON schema encode the formulae. Chapter 3 reports current Symbolica evaluator timings from the example scripts.

Contents

1	Theoretical Construction	2
1.1	Framing, Notation, LTD, and CFF	2
1.2	Repeated Propagators Without Numerator Derivatives	3
1.2.a	The Fully Generic Formula	3
1.2.b	The Box with a Triple Repeated Propagator	4
1.3	Higher Energy Powers in Numerators	5
1.3.1.a	Quadratic Bounds	5
1.3.1.b	One-Loop Box with One Quadratic Edge	6
1.3.2.a	Bounds Above Quadratic	6
1.3.2.b	One-Loop Box with One Cubic Edge	6
2	Python Implementation and JSON Format	8
2.1	Module Structure	8
2.2	Oriented JSON Semantics	8
2.3	Surface Cache	9
2.4	Orientation Labels and Variants	9
2.5	Command-Line Workflow	9
2.6	Symbolica Evaluators	9
3	Performance	10
3.1	Benchmark Setup	10
3.2	Repeated Box	10
3.3	One-Loop 10-Gon	10
3.4	Five-Loop Four-External Benchmarks	11

Chapter 1

Theoretical Construction

1.1 Framing, Notation, LTD, and CFF

Let $\mathcal{G}$ be a connected scalar graph with L loop momenta and internal edge set E . A loop-momentum basis (LMB) is denoted by $\mathcal{B} = (k_1, \dots, k_L)$. For an edge $e \in E$,

$$q_e(k, p) = \sum_{r=1}^L \rho_{er} k_r + \sum_{a=1}^{n_{\text{ext}}} \eta_{ea} p_a, \quad \rho_{er}, \eta_{ea} \in \mathbb{Z}, \quad (1.1)$$

and

$$q_e^0(k^0, p^0) = \rho_e \cdot k^0 + \eta_e \cdot p^0, \quad \omega_e(\vec{k}, p, m) = \sqrt{\vec{q}_e^2 + m_e^2}. \quad (1.2)$$

The Feynman denominator is

$$D_e = (q_e^0)^2 - \omega_e^2 + i0. \quad (1.3)$$

For a numerator N in EMR variables, the D -dimensional input integral is

$$I_{\mathcal{G}, \mathcal{B}}[N] = \int_{\mathbb{R}^{4L}} \prod_{r=1}^L \frac{d^4 k_r}{(2\pi)^4} N(\{q_e(k, p)\}_{e \in E}) \prod_{e \in E} D_e^{-a_e}. \quad (1.4)$$

The powers a_e are normally one; repeated propagators are encoded below as explicit repeated edges or, equivalently, as collapsed channels with powers.

LTD dependency data. The LTD construction is an algorithmic residue expansion of (1.4) after choosing

- the graph $\mathcal{G}$,
- the loop-momentum basis $\mathcal{B}$,
- a contour-closure prescription κ , i.e. the side on which each loop-energy integration is closed.

We denote the resulting spatial integrand by

$$I_{\text{LTD}}^{(3)}[\mathcal{G}, \mathcal{B}, \kappa; N](\vec{k}; p, m) = \sum_{\mathfrak{o} \in \mathcal{O}_{\text{LTD}}(\mathcal{G}, \mathcal{B}, \kappa)} N\left(Q_{\mathfrak{o}}^{\text{LTD}}(\vec{k}; p, m)\right) W_{\mathfrak{o}}^{\text{LTD}}(\vec{k}; p, m). \quad (1.5)$$

The map $Q_{\mathfrak{o}}^{\text{LTD}}$ gives every EMR edge energy after solving the cut equations. The weight $W_{\mathfrak{o}}^{\text{LTD}}$ is a product of cut half-edge factors and LTD dual surfaces. In this note we do not rederive LTD; we use the multiloop algorithmic construction of Ref. [1].

CFF dependency data. For CFF/causal representations we use the graph as the primary input and assume the canonical source/sink contraction ordering used by the implementation. Thus

$$I_{\text{CFF}}^{(3)}[\mathcal{G}; N](\vec{k}; p, m) = \sum_{\mathfrak{o} \in \mathcal{O}_{\text{CFF}}(\mathcal{G})} N\left(Q_{\mathfrak{o}}^{\text{CFF}}(\vec{k}; p, m)\right) W_{\mathfrak{o}}^{\text{CFF}}(\vec{k}; p, m). \quad (1.6)$$

The CFF weights use causal E -surfaces in the denominator. The construction is the one of Refs. [2, 3]. The standard black-box numerator assumption is that N is at most affine in each EMR edge energy q_e^0 .

Surface notation. For any integer vector $c_e \in \{-1, 0, 1\}$ and external-energy shift Δ , define the signed causal surface

$$\eta(c, \Delta) = \sum_e c_e \omega_e + \Delta. \quad (1.7)$$

An E -surface has all non-zero internal-energy coefficients with the same orientation after the canonical CFF construction. An H -surface is a more general signed hyperboloid surface. Pure CFF denominator factors use only E -surfaces; LTD and hybrid terms may use H -surfaces when the residue geometry requires them. Numerator-side surfaces are allowed in the numerator and do not introduce denominator singularities.

1.2 Repeated Propagators Without Numerator Derivatives

Repeated propagators correspond to groups

$$R_g = \{r_{g,1}, \dots, r_{g,\nu_g}\}, \quad \nu_g \geq 2, \quad (1.8)$$

whose members have the same canonical routing and mass, possibly with reversed edge orientation. Let Q_g^0 be the canonical repeated-channel energy and ω_g the common on-shell energy. The purpose of the hybrid construction is to evaluate the repeated-channel residues without applying derivatives to the black-box numerator.

1.2.a The Fully Generic Formula

Start from

$$\begin{aligned} I_{\mathcal{G},B}[N] &= \int \prod_{r=1}^L \frac{d^4 k_r}{(2\pi)^4} N(\{q_e(k, p)\}) \prod_{s \in S} D_s^{-1} \prod_{g \in G_{\text{rep}}} \prod_{a=1}^{\nu_g} D_{r_{g,a}}^{-1} \\ &= \int \prod_{r=1}^L \frac{d^4 k_r}{(2\pi)^4} N(\{q_e(k, p)\}) \prod_{s \in S} D_s^{-1} \prod_{g \in G_{\text{rep}}} D_g^{-\nu_g}. \end{aligned} \quad (1.9)$$

The second equality is the collapsed denominator identity. It is only a denominator identity: the numerator still has one EMR argument for every original edge.

Let $\bar{E} = S \sqcup G_{\text{rep}}$ be the collapsed edge set and assign powers

$$n_{\bar{e}} = \begin{cases} 1, & \bar{e} = s \in S, \\ \nu_g, & \bar{e} = g \in G_{\text{rep}}. \end{cases}$$

For every rank- L collapsed cut basis $B \subset \bar{E}$ and pole-sign vector σ_B , solve the cut equations

$$Q_b^0(k^0, p^0) = \sigma_b \omega_b, \quad b \in B, \quad (1.10)$$

for the loop energies:

$$k^0 = K_{B, \sigma_B}(\vec{k}; p, m). \quad (1.11)$$

The direct residue theorem gives the strict equality

$$I_{\mathcal{G},\mathcal{B}}[N] = \int \prod_{r=1}^L \frac{d^3 k_r}{(2\pi)^3} \sum_{B,\sigma_B} C_{B,\sigma_B} \left[\prod_{b \in B} \frac{1}{(n_b - 1)!} \left(\frac{\partial}{\partial Q_b^0} \right)^{n_b - 1} \frac{N(\{q_e^0(k^0, p^0), \vec{q}_e\})}{\prod_{b \in B} (Q_b^0 + \sigma_b \omega_b)^{n_b} \prod_{\bar{e} \notin B} D_{\bar{e}}^{n_{\bar{e}}}} \right]_{k^0 = K_{B,\sigma_B}}. \quad (1.12)$$

The constant C_{B,σ_B} contains the orientation sign and the Jacobian of the chosen energy basis.

Equation (1.12) is not yet acceptable when a powered channel is cut, because the derivatives act on N . Define the set of active original edges

$$\mathcal{A}_B = \{e \in E : q_e^0 \text{ depends on } Q_b^0 \text{ for some } b \in B \text{ with } n_b > 1\}. \quad (1.13)$$

For the affine EMR class, insert the exact interpolation identity

$$N(\{q_e^0\}) = \sum_{\theta \in \{\pm 1\}^{\mathcal{A}_B}} N(\{q_{e,\theta}^0\}) \prod_{e \in \mathcal{A}_B} \frac{1 + \theta_e q_e^0 / \omega_e}{2}, \quad (1.14)$$

with $q_{e,\theta}^0 = \theta_e \omega_e$ for active edges and the ordinary affine value for inactive edges. In the physical CFF-localized form, the same interpolation is organized as a finite sum over repeated-copy sign orbits obtained by solving the localizing delta constraints before taking the equal-energy limit. Both forms are exact for the affine EMR numerator class.

Substituting (1.14) into (1.12) gives

$$I_{\mathcal{G},\mathcal{B}}[N] = \int \prod_{r=1}^L \frac{d^3 k_r}{(2\pi)^3} I_{\mathcal{G},\mathcal{B},\kappa}^{\text{HybridLTD}}[N](\vec{k}; p, m), \quad (1.15)$$

$$I_{\mathcal{G},\mathcal{B},\kappa}^{\text{HybridLTD}}[N] = \sum_{\mathfrak{o} \in \mathcal{O}_H(\mathcal{G},\mathcal{B},\kappa)} N(Q_{\mathfrak{o}}^H(\vec{k}; p, m)) W_{\mathfrak{o}}^H(\vec{k}; p, m). \quad (1.16)$$

Here one orientation is

$$\mathfrak{o} = (B, \sigma_B, \theta, \mu),$$

where μ labels one algebraic term from expanding the known derivative in (1.12). The numerator is never differentiated; it is only sampled at $Q_{\mathfrak{o}}^H$. The weight is a finite product

$$W_{\mathfrak{o}}^H = c_{\mathfrak{o}} \prod_{h \in H_{\mathfrak{o}}} \frac{1}{2\omega_h} \prod_{\eta \in D_{\mathfrak{o}}} \eta(\vec{k}; p, m)^{-m_{\mathfrak{o},\eta}} \prod_{\zeta \in N_{\mathfrak{o}}} \zeta(\vec{k}; p, m). \quad (1.17)$$

The denominator factors $D_{\mathfrak{o}}$ are cached E - or H -surfaces; the numerator factors $N_{\mathfrak{o}}$ are known surface factors produced by interpolation and contact reductions. If no repeated channel is present, then $n_b = 1$ for all cuts, $\mathcal{A}_B = \emptyset$, and (1.16) collapses exactly to the LTD representation with the same $(\mathcal{G}, \mathcal{B}, \kappa)$.

1.2.b The Box with a Triple Repeated Propagator

Consider the one-loop box routing

$$q_0 = k, \quad q_1 = k + p_1, \quad q_2 = k + p_1 + p_2, \quad Q = k + p_1 + p_2 + p_3,$$

with three repeated copies of Q . The collapsed denominator is

$$D_0 D_1 D_2 D_Q^3.$$

Write $\Delta_i = p_1^0 + \dots + p_i^0$ and define the causal surfaces

$$E_{ij}^{\sigma_i \sigma_j} = \sigma_i \omega_i + \sigma_j \omega_j + \Delta_i - \Delta_j, \quad (1.18)$$

$$E_{iQ}^{\sigma_i \sigma_Q} = \sigma_i \omega_i + \sigma_Q \omega_Q + \Delta_i - \Delta_3. \quad (1.19)$$

The three simple-cut sectors are

$$I_i^{\text{simple}} = - \frac{N(q_0^0, q_1^0, q_2^0, Q^0, Q^0, Q^0)_{k^0 = -\Delta_i + \omega_i}}{2\omega_i E_{ij}^{++} E_{i\ell}^{++} (E_{iQ}^{++})^3}, \quad \{i, j, \ell\} = \{0, 1, 2\}. \quad (1.20)$$

The signs shown correspond to the upper on-shell pole convention used in the examples; the opposite closure is obtained by replacing every displayed $+\omega$ by the corresponding signed on-shell energy.

For the repeated cut, set $Q^0 = \sigma \omega_Q$ and

$$k^0 = \sigma \omega_Q - \Delta_3.$$

Let $\theta = (\theta_3, \theta_4, \theta_5)$ be the sample signs for the three repeated EMR copies. The numerator call is

$$N_{\sigma, \theta} = N(\sigma \omega_Q - \Delta_3, \sigma \omega_Q - \Delta_2, \sigma \omega_Q - \Delta_1, \theta_3 \omega_Q, \theta_4 \omega_Q, \theta_5 \omega_Q). \quad (1.21)$$

The repeated-cut contribution has the finite form

$$I_Q^{\text{rep}} = \sum_{\sigma=\pm 1} \sum_{\theta \in \{\pm 1\}^3} N_{\sigma, \theta} \sum_{\mu} c_{\sigma, \theta, \mu} \frac{\prod_{\zeta \in N_{\sigma, \theta, \mu}} \zeta}{(2\omega_Q)^{3+r_\mu} \prod_{i=0}^2 (E_{Q_i}^{\sigma_i})^{m_{\mu i}}}. \quad (1.22)$$

All objects in (1.22) are explicit functions of external energies and on-shell energies. The coefficients $c_{\sigma, \theta, \mu}$, powers $r_\mu, m_{\mu i}$, and numerator-side surface list $N_{\sigma, \theta, \mu}$ are the expansion of the second-order confluent residue. In the JSON they appear as rational variant prefactors, repeated half-edge entries for $2\omega_Q$, and repeated surface ids. The final boxed formula is

$$I_{\square(3)}^{\text{HybridLTD}} = I_0^{\text{simple}} + I_1^{\text{simple}} + I_2^{\text{simple}} + I_Q^{\text{rep}}.$$

1.3 Higher Energy Powers in Numerators

The CFF construction is naturally compatible with numerators affine in each EMR edge energy. When a user supplies a bound d_e for the highest power of each edge energy in N , the implementation first checks loop-energy UV convergence. In every loop-energy direction v , the degree of the numerator minus twice the number of denominators depending on that direction must be strictly smaller than -1 . If this check fails, finite pole data are not enough and the command refuses to build a representation.

1.3.1.a Quadratic Bounds

Let $x = q_e^0$, $\omega = \omega_e$, and suppose N has degree at most two in x . There is an exact decomposition

$$N(x) = R_e N(x) + (x^2 - \omega^2) C_e N, \quad (1.23)$$

where the pole-safe affine remainder is

$$R_e N(x) = \frac{x + \omega}{2\omega} N(+\omega) + \frac{\omega - x}{2\omega} N(-\omega), \quad (1.24)$$

and the contact quotient is

$$C_e N = \frac{N(+\omega) - 2N(0) + N(-\omega)}{2\omega^2}. \quad (1.25)$$

Therefore

$$\frac{N}{D_e \prod_{f \neq e} D_f} = \frac{R_e N}{D_e \prod_{f \neq e} D_f} + \frac{C_e N}{\prod_{f \neq e} D_f}. \quad (1.26)$$

The first term is affine in q_e^0 and can be handled by ordinary CFF. The second term is a pinched graph with edge e removed, also handled by CFF. Consequently the pure CFF denominator remains E -surface only. Known factors such as $x + \omega$, $\omega - x$, and $1/\omega^2$ are stored as numerator-side surface factors or half-edge powers; they are not denominator surfaces. In hybrid mode the same R/C reduction is applied before the remaining repeated-channel confluent residue is expanded, so the black-box numerator is still sampled only at explicit energies.

1.3.1.b One-Loop Box with One Quadratic Edge

For a non-repeated one-loop box with denominators $D_0 D_1 D_2 D_3$, assume the only non-affine dependence is $d_0 = 2$. Equations (1.23)–(1.26) give

$$I_{\square}^{\text{CFF}}[N] = I_{\text{CFF}}[\square; R_0 N] + I_{\text{CFF}}[\square/0; C_0 N], \quad (1.27)$$

where $\square/0$ is the triangle obtained by deleting propagator 0. More explicitly,

$$I_{\text{CFF}}[\square; R_0 N] = \sum_{\mathfrak{o} \in \mathcal{O}_{\text{CFF}}(\square)} \left[\frac{q_0^0(\mathfrak{o}) + \omega_0}{2\omega_0} N(q_0^0 = +\omega_0) + \frac{\omega_0 - q_0^0(\mathfrak{o})}{2\omega_0} N(q_0^0 = -\omega_0) \right] W_{\mathfrak{o}}^{\text{CFF}}, \quad (1.28)$$

$$I_{\text{CFF}}[\square/0; C_0 N] = \sum_{\mathfrak{t} \in \mathcal{O}_{\text{CFF}}(\square/0)} \frac{N(+\omega_0) - 2N(0) + N(-\omega_0)}{2\omega_0^2} W_{\mathfrak{t}}^{\text{CFF}}. \quad (1.29)$$

Both $W_{\mathfrak{o}}^{\text{CFF}}$ and $W_{\mathfrak{t}}^{\text{CFF}}$ contain only E -surface denominators.

1.3.2.a Bounds Above Quadratic

For $d_e > 2$, let $a_1, \dots, a_{d_e-1}$ be non-pole interpolation nodes with $a_j \neq \pm 1$, and write $x = q_e^0/\omega_e$. If $L_j(x)$ is the Lagrange basis on these nodes, the contact quotient is reconstructed from black-box samples by

$$C_e N(q_e^0) = \sum_j \frac{L_j(q_e^0/\omega_e)}{(a_j^2 - 1)\omega_e^2} \left[N(a_j \omega_e) - \frac{a_j + 1}{2} N(+\omega_e) - \frac{1 - a_j}{2} N(-\omega_e) \right]. \quad (1.30)$$

The remainder part is still the pole interpolation at $\pm\omega_e$. The new feature is that $C_e N$ is now a known polynomial in the remaining energy variables. Those known factors must be carried as numerator-side surfaces and, when their degree is still too high for CFF, recursively reduced by the same finite-pole division. This is why higher powers require E - and H -surface factors in the numerator even though pure CFF denominator trees remain E -surface only. Hybrid mode uses the same bounded-degree contact algebra before or inside each repeated-channel residue sector.

1.3.2.b One-Loop Box with One Cubic Edge

For $N(x) = x^3 M$, with M independent of $x = q_0^0$,

$$x^3 = \omega_0^2 x + x(x^2 - \omega_0^2).$$

Thus

$$\frac{x^3 M}{D_0 D_1 D_2 D_3} = \frac{\omega_0^2 x M}{D_0 D_1 D_2 D_3} + \frac{x M}{D_1 D_2 D_3}. \quad (1.31)$$

The first term has an affine numerator on the box. The second term is a pinched triangle with a known numerator-side linear energy factor x . After the triangle CFF construction samples its numerator map, x becomes a cached surface factor in the numerator. For multiloop examples the same mechanism can produce numerator-side H -surfaces because a pinched lower-sector energy can be a signed difference of on-shell energies. These numerator factors are regular polynomial factors, not denominator singularities.

Chapter 2

Python Implementation and JSON Format

2.1 Module Structure

The implementation is organized around an evaluator-complete intermediate format.

- `graph_io.py` parses DOT graphs, extracts momentum signatures, detects repeated channels, and builds split-mass reference graphs.
- `graph_signatures.py` implements the `graph_from_signatures` command, which turns products such as `prop(k1+p1,m)` into DOT graphs.
- `orientation_bundle.py` constructs LTD, CFF, hybrid, and bounded-degree contact bundles. This is where repeated channels are collapsed, active numerator maps are enumerated, and CFF lower minors are generated.
- `structure.py` serializes bundles to schema-v6 JSON and contains the arbitrary-precision Python evaluator.
- `pretty.py` renders the JSON in a human-readable form.
- `symbolica_eval.py` compiles a fixed-numerator JSON expression to Symbolica eager and assembly-backed evaluators.

2.2 Oriented JSON Semantics

The top-level JSON contains graph metadata, a surface cache, and a list of orientations. The central invariant is:

one orientation corresponds to one unique EMR edge-energy numerator map.

All denominator terms sharing that numerator call are stored as **variants**. Each variant has its own rational prefactor, half-edge list, numerator-surface list, and factorized denominator tree.

The evaluator uses only the JSON plus the DOT graph. It does not reconstruct a hidden LTD or CFF object. A variant contributes

$$\text{pref} \prod_{e \in \text{half_edges}} (2\omega_e)^{-1} \prod_{s \in \text{num_surfaces}} \eta_s \prod_{t \in \text{denominator tree}} \eta_t^{-1}$$

times the numerator sampled at the orientation's `edge_q0` map.

2.3 Surface Cache

Each surface cache entry records a class, coefficients of internal on-shell energies, and external-energy shifts. The class is printed as **e** or **h**. If a surface appears only in numerator factors, the entry carries `numerator_only=true` and pretty printing displays **(e)** or **(h)**. If a numerator-side surface is identical to an existing denominator surface, the same id is reused.

2.4 Orientation Labels and Variants

Compact labels use $+$, $-$, 0 , and x :

- $+$: the numerator samples that edge at $+\omega_e$;
- $-$: the numerator samples that edge at $-\omega_e$;
- 0 : the numerator samples that edge at zero energy;
- x : the edge has a non-trivial affine map, printed by `-show-details`.

The physical origin of a contribution, such as CFF, LTD, pinched contacts, or hybrid basis/interpolation data, is variant metadata rather than part of the orientation identity. This is what keeps the number of numerator calls minimal.

2.5 Command-Line Workflow

The main commands are:

```
python3 hybrid3d.py validate --dot examples/graphs/box_pow3.dot
python3 hybrid3d.py build --family hybrid --dot examples/graphs/box_pow3.dot --pretty
python3 hybrid3d.py test --dot examples/graphs/box_pow3.dot
python3 hybrid3d.py test-cff-ltd --dot examples/graphs/box.dot --energy-degree-bounds
0:2
python3 hybrid3d.py compile --orientation-json demo.json --dot graph.dot --numerator-
expr "1"
python3 hybrid3d.py evaluate --orientation-json demo.json --dot graph.dot --evaluator-
backend symbolica
```

The repeated-propagator three-way test compares pure CFF, hybrid, and the split-mass LTD limiting proxy. The ordinary LTD formula on an unsplit repeated graph is not a valid physics comparison because numerator derivatives are not included there.

2.6 Symbolica Evaluators

The `compile` command accepts one JSON representation and one fixed numerator expression. It reconstructs the JSON expression symbolically, introducing structured functions such as `OSE(qx,qy,qz,m)`, `etaE(i)`, `etaH(i)`, and tagged edge/loop/external momentum components. The resulting Symbolica evaluator is saved both as a shared library and as eager evaluator state. The JSON is updated with relative paths and fingerprints tying the evaluator to the graph, numerator, value type, and orientation structure.

The `evaluate` command can use:

```
builtin,    symbolica_compiled,  symbolica_eager,  symbolica_eager_symjit.
```

Profiling performs ten calls of a batch evaluator. The default batch size is 100, so a profile line reports one thousand repeated samples of the same input kinematics.

Chapter 3

Performance

3.1 Benchmark Setup

The numbers below were generated on April 26, 2026 using the scripts in `examples/scripts`. Each timing is the per-sample time reported by `evaluate -profiling 100`, i.e. ten batch calls of size 100. The 15-external one-loop benchmark is intentionally omitted here: the LTD build is small, but the pure CFF one-loop representation has $2^{15} - 2 = 32766$ sign sectors and the CFF build is not practical for routine documentation runs.

3.2 Repeated Box

The repeated box script `examples/scripts/box_pow3_cli_showcase.sh` compares hybrid and pure CFF for the six-edge graph with a triple repeated propagator. The numerator is `dot(edges[0], ext[0]) + edges[3][0]`.

label	backend	orientations	maps	value type	per sample
hybrid	compiled	17	17	real	$0.0955\ \mu s$
hybrid	eager	17	17	real	$0.521\ \mu s$
hybrid	eager+symjit	17	17	real	$0.104\ \mu s$
hybrid	compiled	17	17	complex	$0.271\ \mu s$
hybrid	eager	17	17	complex	$0.725\ \mu s$
hybrid	eager+symjit	17	17	complex	$0.0989\ \mu s$
CFF	compiled	62	62	real	$0.162\ \mu s$
CFF	eager	62	62	real	$1.44\ \mu s$
CFF	eager+symjit	62	62	real	$0.142\ \mu s$
CFF	compiled	62	62	complex	$0.466\ \mu s$
CFF	eager	62	62	complex	$2.03\ \mu s$
CFF	eager+symjit	62	62	complex	$0.136\ \mu s$

The hybrid representation has fewer numerator maps than CFF, which explains the faster compiled real evaluator. Complex compiled evaluation is slower because the generated function propagates complex arithmetic even though the input kinematics are real. Symjit eager evaluation is competitive for this small expression because the batch is small and the expression has only 17 or 62 maps.

3.3 One-Loop 10-Gon

The script `examples/scripts/one_loop_10_external_runtime_compare.sh` compares numerator-free LTD and CFF on a true one-loop 10-gon.

family	backend	orientations	maps	per sample
LTD	compiled	10	10	$0.234\ \mu s$
LTD	eager	10	10	$0.852\ \mu s$
LTD	eager+symjit	10	10	$0.147\ \mu s$
CFF	compiled	1022	1022	$3.96\ \mu s$
CFF	eager	1022	1022	$36.6\ \mu s$
CFF	eager+symjit	1022	1022	$2.97\ \mu s$

The result is expected: the one-loop CFF orientation count grows like $2^n - 2$, while LTD has only one cut per internal edge. The CFF expression is still fast in absolute terms after compilation, but it is much larger.

3.4 Five-Loop Four-External Benchmarks

The script `examples/scripts/five_loop_symbolica_runtime_compare.sh` compares three numerator choices on a non-repeated five-loop graph and on the repeated five-loop graph. The three numerator choices are 1, one edge-external dot product per internal edge, and the same numerator with three selected edge terms squared.

case	family	maps	compiled	eager	eager+symjit
no repeats, $N = 1$	LTD	137	0.326	1.88	0.250
no repeats, $N = 1$	CFF	930	1.26	12.3	0.892
no repeats, $N = 1$	hybrid	137	0.333	1.87	0.246
no repeats, linear	LTD	137	0.481	3.29	0.379
no repeats, linear	CFF	930	1.96	17.5	1.35
no repeats, linear	hybrid	137	0.484	3.24	0.391
no repeats, squared	LTD	137	0.493	3.45	1.73
no repeats, squared	CFF	4692	18.7	45.9	13.2
no repeats, squared	hybrid	137	0.520	3.31	0.390

case	family	maps	compiled	eager	eager+symjit
repeated, $N = 1$	LTD	137	0.334	1.90	0.272
repeated, $N = 1$	CFF	930	1.26	12.5	0.899
repeated, $N = 1$	hybrid	630	0.823	4.88	0.579
repeated, linear	LTD	137	0.481	3.44	0.408
repeated, linear	CFF	930	1.88	17.4	1.32
repeated, linear	hybrid	630	21.4	37.8	12.4
repeated, squared	LTD	137	0.491	3.53	0.387
repeated, squared	CFF	115	1.89	9.49	1.16
repeated, squared	hybrid	115	1.84	9.47	1.11

For non-repeated graphs the hybrid family collapses to the LTD structure, so their orientation and map counts agree. CFF is slower because it has more causal sign sectors. The squared CFF no-repeat case is much larger because the bounded-degree contact completion adds pinched CFF minors.

For repeated graphs, the LTD rows are included only as runtime references. The ordinary unsplit LTD expression does not include numerator derivatives for raised propagators; the correct comparison is CFF versus hybrid versus the split-mass LTD limiting test. The repeated unit numerator shows the expected middle ground: hybrid has fewer maps than CFF but more

than LTD. The repeated linear case is slower for hybrid because the confluent interpolation creates heavier denominator variants per numerator map. In the squared repeated case, the bounded-degree reduction compresses both CFF and hybrid to 115 maps and the two timings become nearly identical.

Bibliography

- [1] J. J. Aguilera-Verdugo et al., algorithmic multiloop LTD construction, arXiv:1906.06138.
- [2] Z. Capatti et al., causal/CFF representation construction, arXiv:2211.09653.
- [3] Z. Capatti et al., further CFF representation and box examples, arXiv:2311.14374.
- [4] cLTD and gammaLoop implementations, <https://github.com/alpha100p/cLTD> and <https://github.com/alpha100p/gammaloop>.